

Telecom CDR ETL Pipeline — Hackathon

Module 3: Data Loading & Database Design

Assigned to: Member 3

System Architecture



Daily CDR Source Expr/Filter/ stg_cdr Revenue CSV File Qualifier Lookup final_cdr KPIs

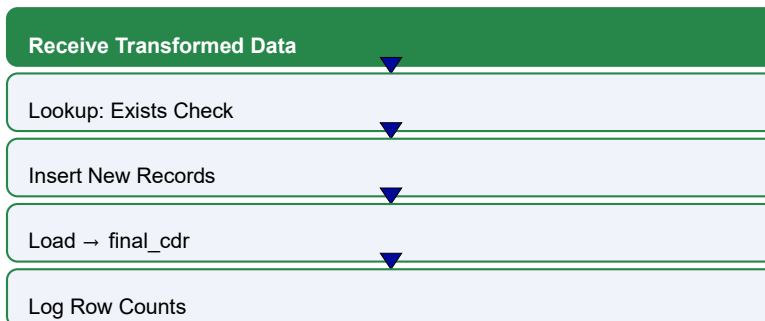
Objective

Design and create the final database tables, load the cleaned and transformed CDR data from staging into the final production table ready for KPI reporting.

1. Design Phase

- Design staging table schema: stg_cdr
- Design final table schema: final_cdr with all computed fields
- Plan incremental load strategy (avoid reloading existing records)
- Design Lookup to check if record already exists before insert
- Plan error table: err_cdr for rejected records

2. Workflow



3. Implementation

- Create DB tables: stg_cdr, final_cdr, err_cdr
- Final table columns: CustomerID, CallType, Duration, Revenue, IsInternational, CallDate, TowerID
- Lookup Transformation: check if CustomerID+CallDate already loaded
- Target Designer: configure final_cdr as Informatica target
- Load rejected records into err_cdr with error reason

Stretch Features

- Incremental load (only new records)
- Audit table with load timestamps

Tools

Informatica Target Designer, Lookup Transformation, SQL Database

Final Output final_cdr table fully populated with cleaned, enriched CDR data — ready for KPI queries

Team Collaboration Notes

- Use shared GitHub repository — commit your Informatica mappings and SQL scripts
- Follow naming conventions: stg_cdr, final_cdr, err_cdr
- Your output feeds into: Module 4
- Test your session using Informatica Workflow Monitor before handoff
- Communicate any schema changes with all members immediately